

Accessing and Distributing Large Volumes of NetCDF Data

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Abstract- In this paper, we will discuss how NASA's Oak Ridge National Laboratory Distributed Active Archive Center (ORNL DAAC) is distributing large volumes of 'structured' data using Daily Surface Weather Data and a corresponding Climatological Summaries Dataset (Daymet) as an example.

Keywords- Daymet, ORNL DAAC, Single Pixel Tool, THREDDS.

I. INTRODUCTION

Archived and distributed through the ORNL DAAC, the Daymet (Version 3) [1] data set provides gridded estimates of daily weather parameters for North America, including daily continuous surfaces of minimum and maximum temperature, precipitation occurrence and amount, humidity, shortwave radiation, snow water equivalent, and day length. The daily time step, 1 km x 1 km spatial resolution, and North American spatial extent of the data set makes it a unique and valuable contribution to the scientific, research, and educational communities. Daymet data are available for 1980 through the latest full calendar year and includes the United States, Mexico, Canada, Hawaii, and Puerto Rico. The data are stored and distributed as individual netCDF file format (.nc4) for each parameter. For each spatial region, there are 252 *.nc4 files, that is, one file for each of the seven Daymet parameters for 36 years (1980 -2015). In total, there are 756 files and the total size of the Daymet data is roughly 20 terabytes (TB).

II. IMPLEMENTATION

Daymet Data Distribution - The Daymet data have broad applications over a wide variety of research fields including hydrology, terrestrial vegetation growth models, carbon cycle science, and regional to large scale climate change

analysis. Example- Figure 1 shows the annual precipitation summary map produced using of Daymet data. The access to the Daymet data set is available through a variety of tools (<https://daymet.ornl.gov/tools.html>) and formats allowing a rich resource of daily surface meteorology.

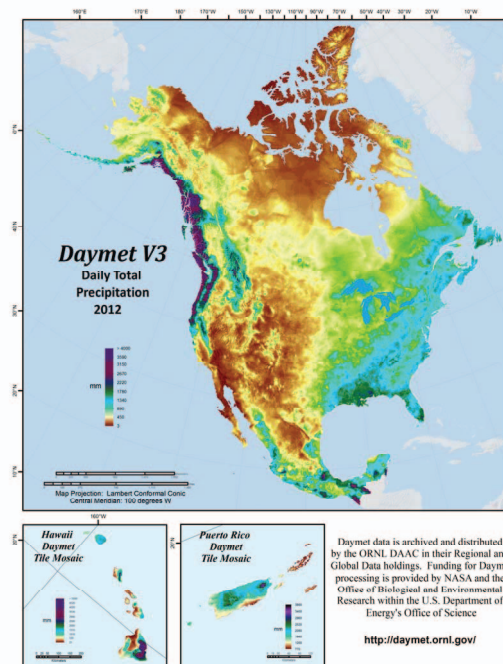


Figure 1. Example of summary map produced using Daymet data. Map shows Daily Annual Total Precipitation in 2012.

(i) The Single Pixel Data Extraction Tool (<https://daymet.ornl.gov/single-pixel/>) is a Java-based tool that allows users to enter a single geographic point by latitude and longitude coordinate in decimal degrees. A routine is executed that translates the geographic coordinates into projected Daymet (x,y) coordinates. These coordinates are used to access the Daymet database containing daily-

interpolated surface weather variables. Daily data from the nearest 1 km x 1 km Daymet grid cell are extracted from the database and formatted as a table with one column for each Daymet variable and one row for each day. All daily data for selected years are returned as a single (long) table, formatted for display in the browser window. The Single Pixel Data Extraction Tool also provides the option to download multiple coordinates programmatically. The Single Pixel Tool uses the netCDF Operator (NCO), which comprise of several command line programs that take netCDF files as input, then operate and output the results to terminal or ascii formats. Example: This command creates an output.nc which contains *time*, *pressure*, and any coordinate variables associated with *time* and *pressure* present in input.nc:

"ncks -C -v time,pressure input.nc output.nc".

(ii) The Daymet Tile Selection Tool provides daily data for each Daymet variable as a separate CF-compliant netCDF file. Data are downloaded in 2-degree x 2-degree tiles for selected years. For researchers who need continuous surfaces of daily meteorologic data over study areas larger than the Single Pixel Extraction provides, Daymet data are distributed as gridded daily data in 2-degree x 2-degree tile subsets. The tile subsets are referenced by TileID, which is assigned within the Daymet algorithm. When downloaded, the daily gridded tile data contain a tarred file of individual netCDF files for each Daymet variable.

(iii) The other popular tool for distributing these large collections of Daymet data is via Thematic Real-time Environmental Distributed Data Services (THREDDS). The ORNL DAAC's THREDDS Data Server (TDS; http://daymet.ornl.gov/thredds_mosaics.html) [2] provides dynamic and customized visualization and access to Daymet time series of North American mosaics. The ORNL DAAC TDS (Figure 2) provides a collection of community, standards-based services for users to visualize and access Daymet data. The Open-source Project for a Network Data Access Protocol (OPeNDAP) provides an efficient, standards-based service to subset Daymet data based on time, space, or variable. The ORNL DAAC TDS also provides other services for data access and visualization, including NetCDF Subset Service, Open Geospatial Consortium (OGC) Web Coverage Service (WCS), and Web Map Service

(WMS). The ORNL DAAC TDS also exposes Daymet's CF-metadata through its ncISO service to facilitate harvesting Daymet metadata records into third party catalogs. The ORNL DAAC TDS approach made it easier for users to use the Daymet data and integrate Daymet data into third party applications and/or data systems (Figure 2).

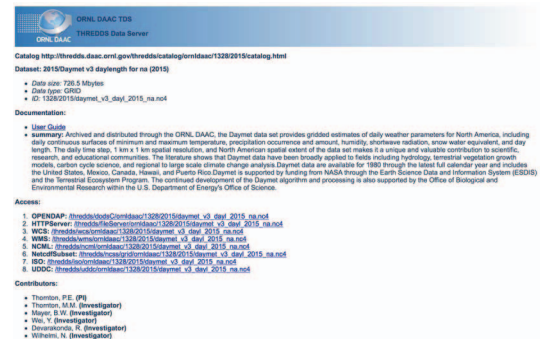


Figure 2. ORNL DAAC THREDDS Data Server (TDS) showing different data access options.

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